



Human Anatomy & Physiology

Project Presentation by

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Objective of our Project

Design a system that mimics the pancreas.

Adjusting insulin and monitoring blood glucose continuously.



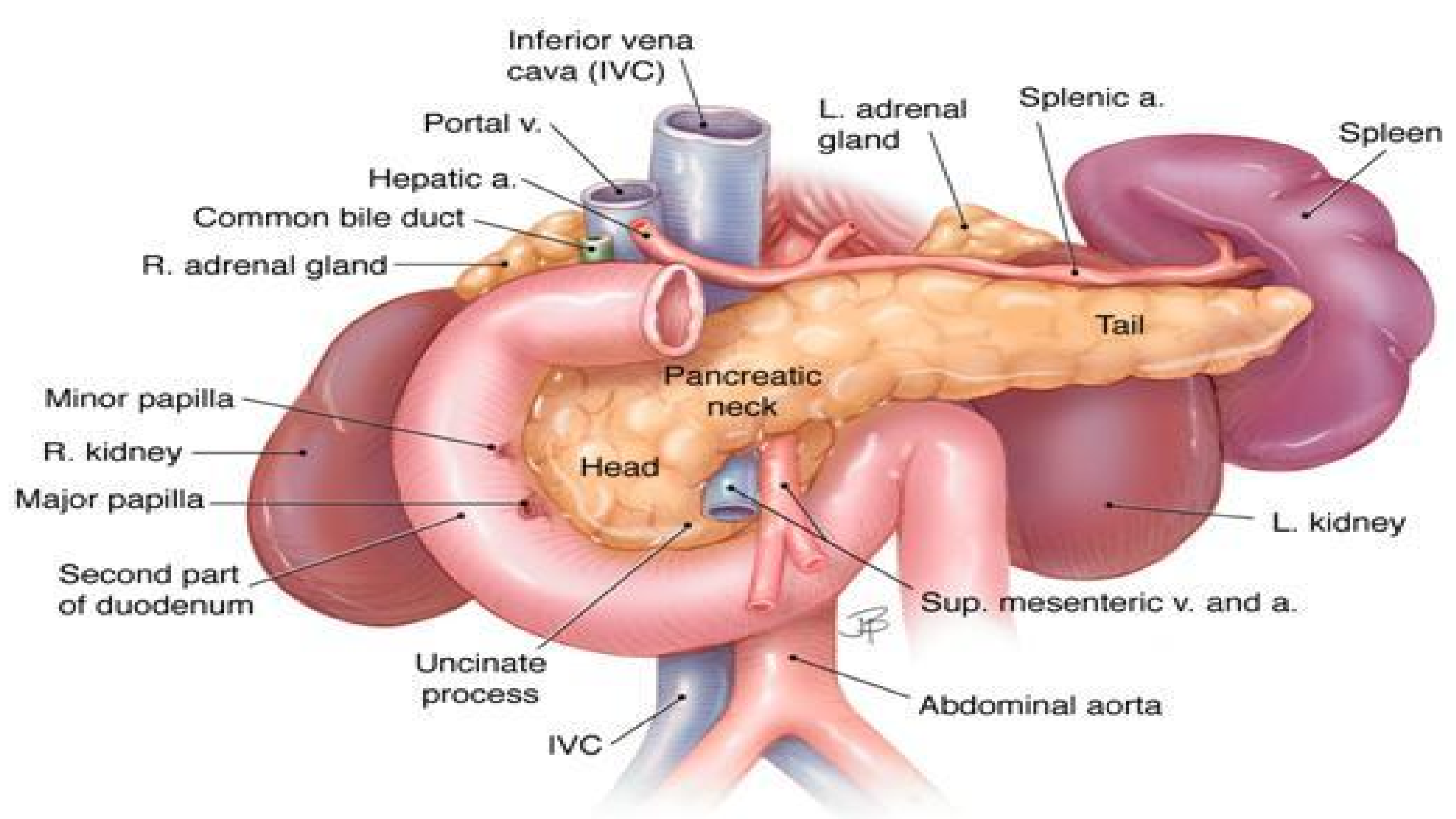
PANCREAS

- ▮ The pancreas is a **glandular organ** with both **endocrine** and **exocrine** functions.
- ▮ Lies **horizontally** in the **upper abdomen**, behind the **stomach**.
- ▮ **Tadpole-like, flat, and elongated.**

Gross Anatomy – Parts of Pancreas

- **Head:** Broadest part; sits in the duodenal curve.
- **Body:** Central, elongated portion.
- **Tail:** Narrow end; extends toward the spleen.





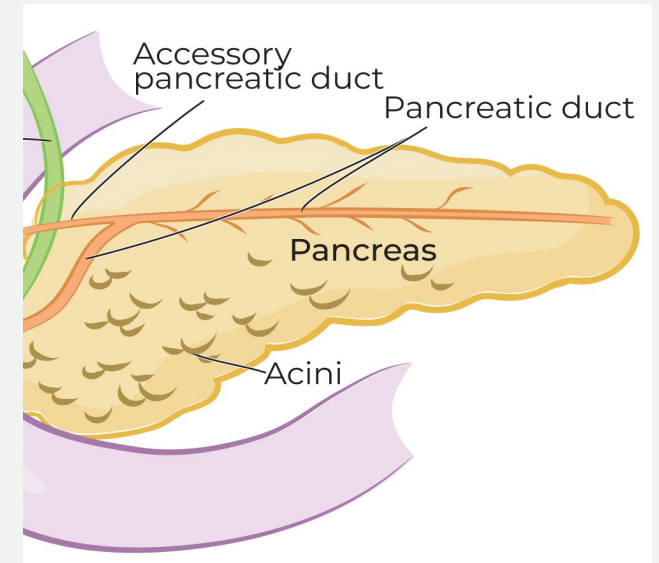
Pancreatic Duct System

Main Pancreatic Duct :

It carries digestive juices from the pancreas to the small intestine through a main opening.

Accessory Pancreatic Duct :

It carries digestive juices through a smaller opening into the small intestine.



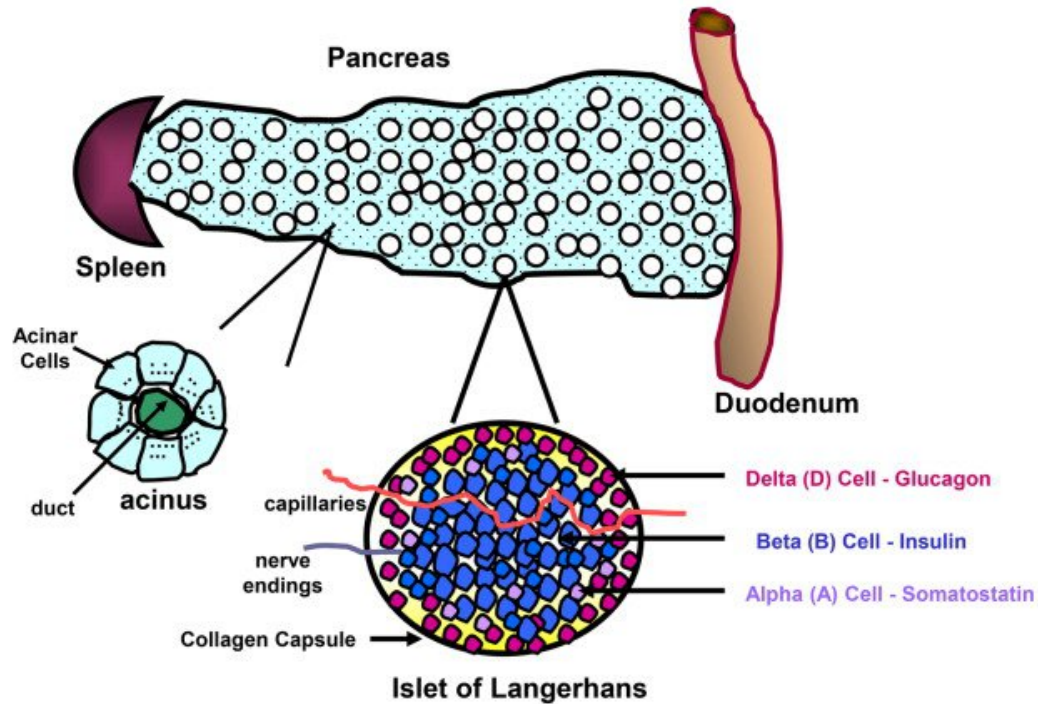
Functional Anatomy of the Pancreas

E x o c r i n e - F u n c t i o n

The pancreas makes digestive juices (like amylase, lipase, and proteases) that help break down food in the small intestine.

E n d o c r i n e - F u n c t i o n

The pancreas has small groups of cells called **Islets of Langerhans** that make hormones:



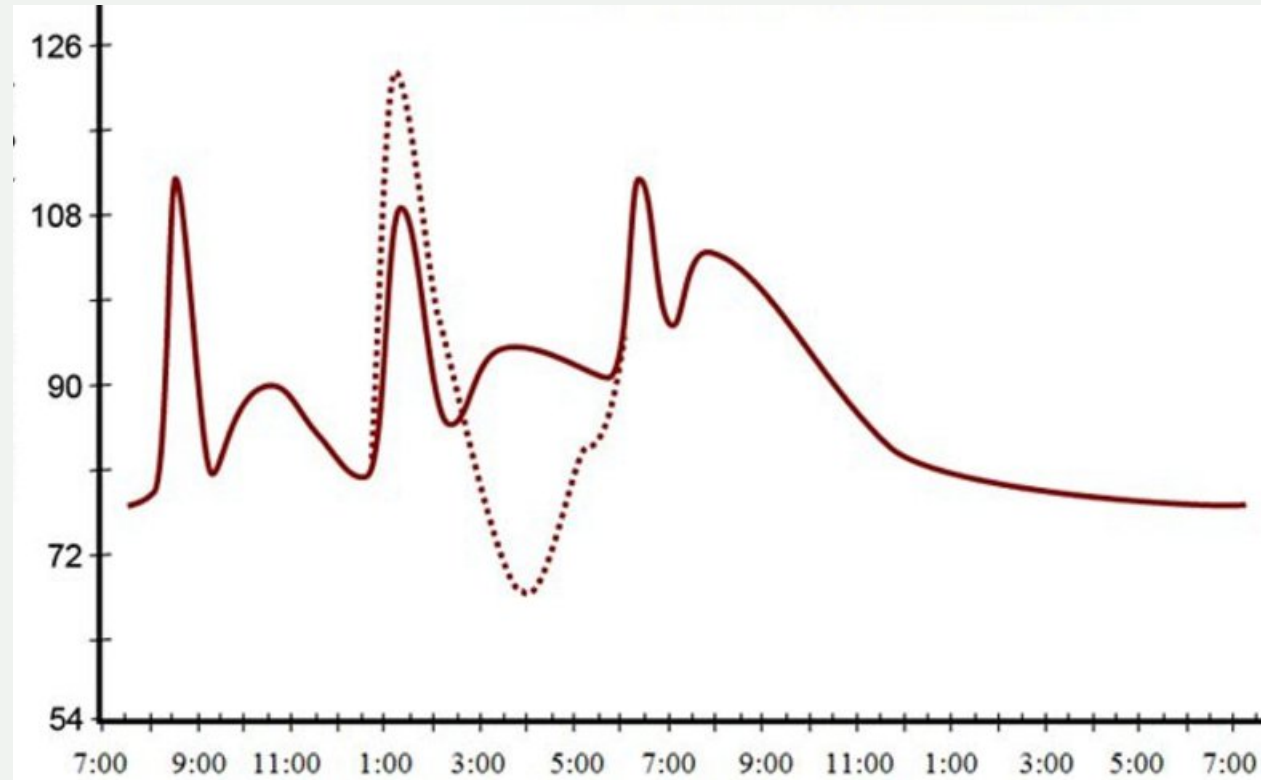
- Alpha cells:** Make **glucagon**, which raises blood sugar when it's too low.

- Beta cells:** Make **insulin**, which lowers blood sugar when it's too high.

- Delta cells:** Make **somatostatin**, which helps control the release of other hormones.

- PP cells:** Make **pancreatic polypeptide**, which helps control hunger and digestion.

Blood glucose level regulation



Production in the Pancreas

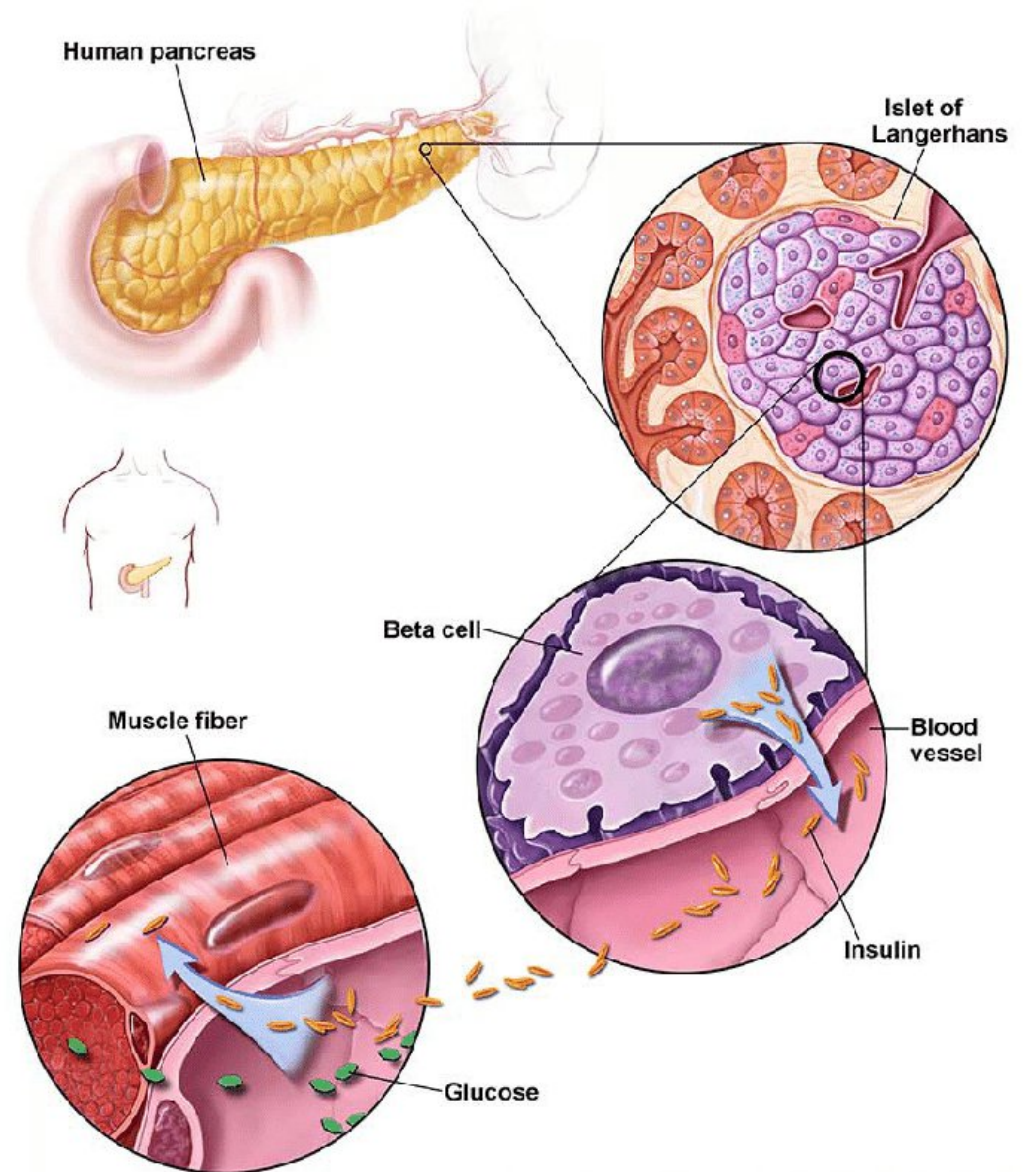
Insulin Production:

Produced by **beta cells** in the **Islets of Langerhans** in the pancreas.

Role of Insulin:

Helps **lower blood sugar** by allowing cells to absorb glucose.

Stores excess glucose in the **liver** for future energy.

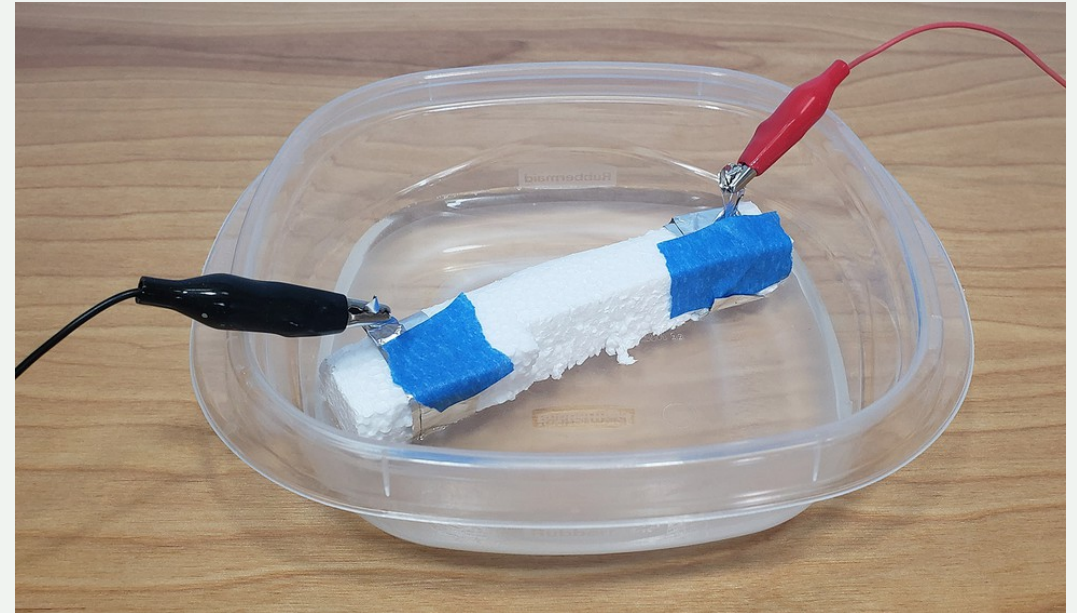


Material

- Breadboard
- Arduino Mega
- N-Channel MOSFET
- 5-V peristaltic Liquid Pump
- 100 k Ω resistor
- Alliga 100 k Ω resistor

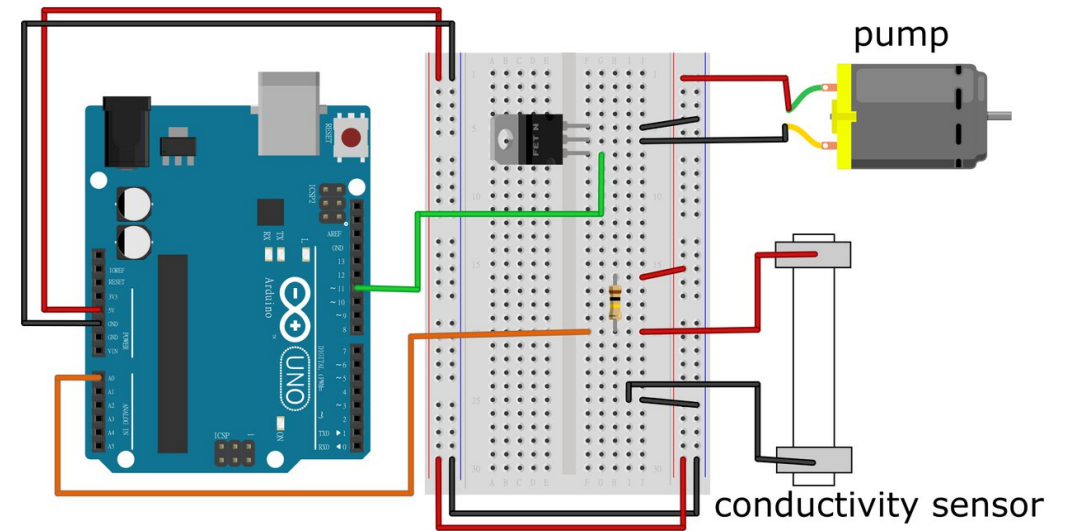
Building the Conductivity Sensor

- ▮ **Materials:** Use two pieces of aluminum foil and something to make it float.
- ▮ **Design:** Place the aluminum foil strips apart so they float in the water.
- ▮ **Function:** Make sure the foil stays at the same level in the water for steady measurements.
- ▮ **Connection:** Attach clips to each piece of foil.



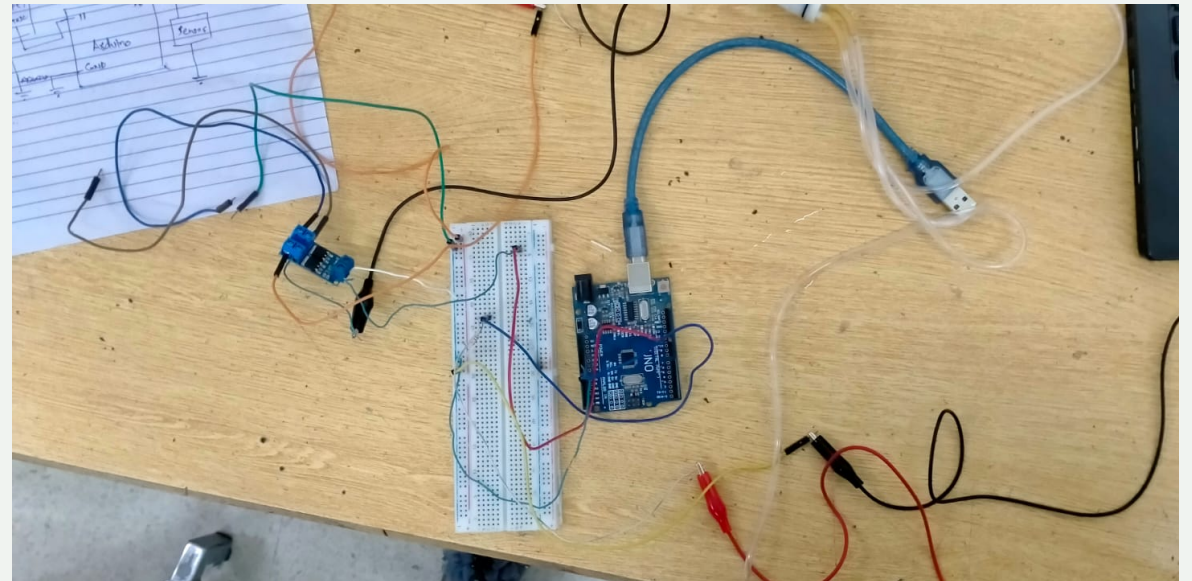
Building the Circuit

- ❑ **Connect Power:** Link 5V from Arduino to the power (+) bus and GND to the ground (-) bus on the breadboard.
- ❑ **MOSFET Placement:** Insert MOSFET on the breadboard with the metal tab facing left. Connect the first pin to Arduino pin 11 and the third pin to the ground bus.
- ❑ **Resistor Setup:** Place the 100 k Ω resistor between the power bus and Arduino analog pin A0.
- ❑ **Sensor Connection:** Connect one end of the sensor to the resistor and the other end to the ground bus.
- ❑ **Pump Connection:** Connect the pump's "+" tab to the power bus and the other tab to the MOSFET's middle pin.

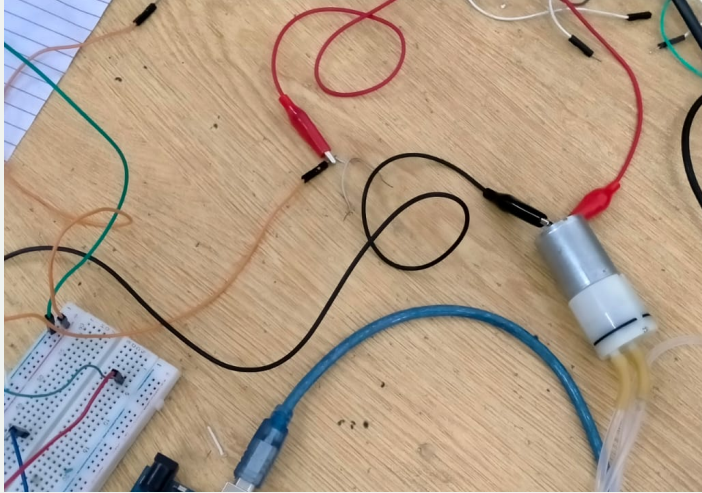


Calibrating the Conductivity Sensor

- ❑ **Prepare Sensor:** Fill a container halfway with distilled water. Place the sensor in the water, ensuring it floats. Adjust the alligator clips if needed.
- ❑ **Disconnect Pump:** Temporarily disconnect the pump to avoid it turning on during calibration.
- ❑ **Power Arduino:** Plug in the USB cable from the Arduino to your computer to power up the circuit. If there's previous code, it will start, but it won't affect calibration.
- ❑ **Using a Multimeter:** Measure DC voltage, connect probes, and convert the reading to an `analogRead` value.
- ❑ **Without Multimeter:** Upload code, open Serial Monitor, and record the average `analogRead` value.



Pump Testing & Final Calibration



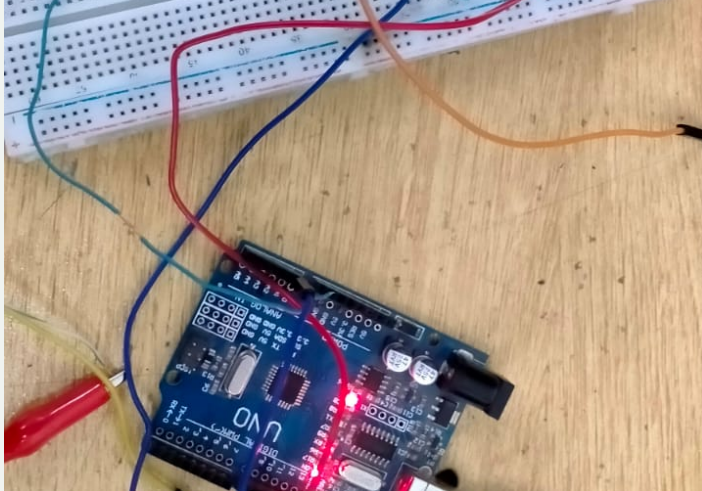
Pump Testing:

Fill a container with tap water, dry and place the sensor in it. Record max and min values for distilled and tap water. The system will simulate blood glucose changes by adding tap water (insulin) when high glucose is detected.

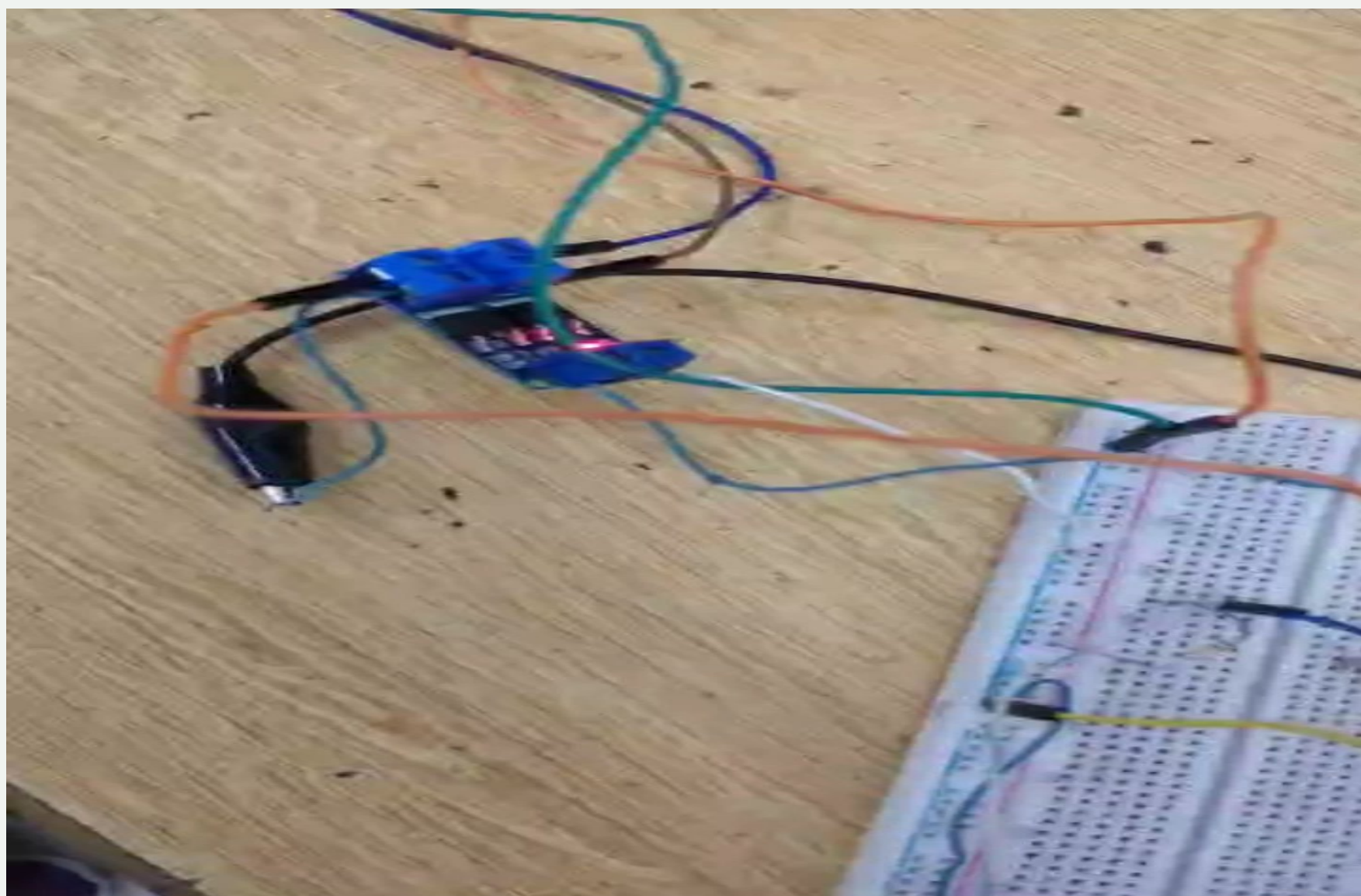
Finalize Calibration:

Test the pump's flow direction and mark it. Set the pump's inlet in tap water and outlet in distilled water. Clean the sensor and return it to distilled water after testing.

Testing & Troubleshooting the Pump and Code



- **Upload & Set Threshold:**
- Choose a threshold value (between 0-1023) for the pump to turn off. Upload the code and open the Serial Monitor.
- **Reconnect Pump:**
- Reconnect the pump's wire to the MOSFET and start pumping. Stir the water gently to mix it.
- **Check Output:**
- Ensure the pump stops when the threshold is reached and the value stabilizes.
- **Troubleshooting:**
- If the pump doesn't start, check the MOSFET wiring.
- If it doesn't stop, check the sensor, resistor, and threshold value.





Thank You